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The Radial Hamiltonian Operator for LiH $X^1\Sigma^+$

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Precise data for rotational and vibration–rotational transitions involving low-lying $X^1\Sigma^+$ vibrational levels of ^{6}LiH, ^{7}LiH, ^{6}LiD, and ^{7}LiD are employed simultaneously in a least-squares procedure for determination of the radial Hamiltonian operator for nuclear motion. Synthetic line positions from eigenvalues obtained by numerical solution of the Schrödinger equation with the fitted Hamiltonian are in agreement with the experimental data to within the measurement accuracies. The equilibrium separation R_e of the Born–Oppenheimer potential is in close agreement with several previous estimates that take account of Born–Oppenheimer breakdown effects. The results obtained for the Born–Oppenheimer potential, as well as the associated radial functions that describe homogeneous and heterogeneous mixing, supercede the set of functions derived by J. F. Ogilvie (*J. Mol. Spectrosc.* **148,** 243–249 (1991)) using an algebraic method, which are now known to be in error. It has not yet been established that eigenvalues of an algebraically determined operator can generate line positions that are in agreement with experimental positions to within the measurement uncertainties. © 1992 Academic Press, Inc.

INTRODUCTION

Extensive spectroscopic data are available for the $X^1\Sigma^+$ ground state of diatomic lithium hydride. Most of these data, which extend to high vibrational levels, are from electronic transitions and are summarized by Chan *et al.* (*1*). The more recent work has been concerned with obtaining highly precise measurements of rotational and of vibration–rotational transitions involving the lower vibrational levels of ^{6}LiH, ^{7}LiH, ^{6}LiD, and ^{7}LiD. Plummer *et al.* (*2*) measured $J = 1 \leftarrow 0$ transitions in ^{6}LiH ($v = 0$) and ^{7}LiH ($v = 0, 1$), and $J = 2 \leftarrow 1$ transitions in ^{6}LiD ($v = 0$) and ^{7}LiD ($v = 0, 1$) with an accuracy of 50 kHz. These data supplemented the earlier data of Pearson and Gordy (*3*) for the $J = 1 \leftarrow 0$ transitions in ^{6}LiD ($v = 0, 1$) and ^{7}LiD ($v = 0, 1$). Fourty-three vibration–rotational transitions in the $\Delta v = 1$ sequences of the four isotopomers have been obtained with an accuracy of 0.001 cm^{-1} by Yamada and Hirota (*4*) using infrared diode laser spectroscopy. Very recently, Maki *et al.* (*5*) obtained a much more extensive collection of rotational and vibration–rotational data using Fourier transform spectroscopy.

LiH is one of the lightest known heteronuclear diatomics, and the recent data (*2–5*) have naturally been of much interest in characterizing breakdown of the Born–Oppenheimer approximation. Plummer *et al.* (*2*) were concerned only with the 10 observed microwave frequencies (for one of which, the $J = 1 \leftarrow 0$ transition in ^{7}LiD ($v = 0$), there is a typographical error in reproducing the original measurement (*3*)). The usual Dunham expansion,

$$E(v, J) = \sum_{kl} Y_{kl}(v + \tfrac{1}{2})^k[J(J + 1)]^l \qquad (1)$$

was modified following Watson's (*6*) treatment of non-adiabatic effects by writing for each isotopomer,

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$$Y_{01} = B_e^{BO} + \Delta^{Li}/M_{Li} + \Delta^{H}/M_H, \quad (2)$$

where B_e^{BO} is the Born–Oppenheimer B_e value, and Δ^{Li}/M_{Li} and Δ^{H}/M_H are corrections arising from each atomic center. Although the least-squares fit employed 7 adjustable parameters with only 10 line positions (3 degrees of freedom), B_e^{BO} was determined precisely, giving $R_e^{BO} = 1.594\,913\,2(6)$ Å. A similar approach was employed by Pearson and Gordy (*3*), except that corrections of the type given by Eq. (2) for Y_{01} were also employed for Y_{02}, Y_{10}, Y_{11}, and Y_{20}. With the submillimeter data of Ref. (*2*) included in the fit, the fitted value of $R_e^{BO} = 1.594\,914\,3(6)$ Å is consistent with that of Plummer *et al.* (*2*) to within the combined uncertainties. Maki *et al.* (*5*) performed a similar analysis with their more extensive data; the value for the isotopically invariant parameter U_{01} obtained in Ref. (*5*) gives $R_e^{BO} = 1.594\,911(1)$ Å, slightly less than, but in agreement with the previous determinations.

Maki *et al.* (*5*) also realized the advantages of fitting the observed data directly with a model potential, rather than to a set of physically unconstrained U_{kl} and Δ_{kl} parameters. However, with only an approximate treatment of Born–Oppenheimer breakdown, a fit of this type using a Dunham potential was not satisfactory; the standard deviation of the fit was over four times that for the fit of the same data to mass-independent Dunham constants and Δ_{kl} parameters. Even for the data for a single isotope, Maki *et al.* were unable to obtain satisfactory fits of an effective Dunham potential, and it was concluded that non-adiabatic mixing was responsible for J-dependent Born–Oppenheimer breakdown effects not included in their model.

Maki *et al.* were not aware of the recent work (*7, 8*) from this laboratory on the ground states of HCl/DCl and HF/DF, for which J-dependent non-adiabatic mixing effects were characterized unequivocally. The model that we have developed is based on Watson's (*9*) more recent and refined treatment of non-adiabatic mixing, and enables the radial Hamiltonian operator for an individual isotope to be written as

$$\mathscr{H}^{eff}(R) = P_R^2/(2\mu_{at}) + U^{eff}(R) + B_e(R_e/R)^2 J(J+1)[1 + q(R)], \quad (3)$$

where μ_{at} is the atomic reduced mass, and the functions $U^{eff}(R)$ and $q(R)$ are defined by

$$U^{eff}(R) = U^{BO}(R) + m_e[\tilde{S}_A(R)/M_A + \tilde{S}_B(R)/M_B] \quad (4)$$

and

$$q(R) = m_e[\tilde{R}_A(R)/M_A + \tilde{R}_B(R)/M_B]. \quad (5)$$

$\tilde{S}_A(R)$, $\tilde{S}_B(R)$, $\tilde{R}_A(R)$ and $\tilde{R}_B(R)$ are correction functions from the two atomic centers A and B that collectively take account of adiabatic and non-adiabatic corrections (and other smaller effects).

In our HF/DF and HCl/DCl work, for which the available data extend close to dissociation, the $U^{BO}(R)$ functions were modeled numerically. However, in more recent work (*10*) on HBr/DBr and HI/DI, we have successfully demonstrated the same approach with $U^{BO}(R)$ specified analytically. The parameters defining $U^{BO}(R)$ and the functions $\tilde{S}_i(R)$ and $\tilde{R}_i(R)$ are obtained by numerical solution of the radial wave equation, with the necessary partial derivatives being calculated using the generalized Hellmann–Feynman theorem (*11*).

The results obtained for the hydrogen halides have been of interest in the development of an alternative algebraic approach. Ogilvie (*12*) has obtained expressions for Dunham Y_{kl} parameters and mass scaling factors Δ_{kl}, and their partial derivatives, in terms of the parameters of the corresponding Dunham-like potential,

$$U(z) = c_0 z^2(1 + \sum_{j=1} c_j z^j), \tag{6}$$

and of the functions $K(z) = \sum h_j z^j$ for the two atomic centers, introduced as a correction for the combined effects of inadequacy of the semiclassical JWKB approximation and Born–Oppenheimer breakdown. The functions are defined in terms of the coordinate $z \equiv 2(R - R_e)/(R + R_e)$ rather than the more familiar coordinate, $x \equiv (R - R_e)/R_e$. However, it is only in a very recent publication (*13*) that the need for functions analogous to $q(R)$ in Eqs. (3) and (5) was considered explicitly. In earlier work (*14*), the derived c_j and h_j parameters necessarily absorbed the J-dependent non-adiabatic effects.

Since the effective potential employed by Ogilvie (*13*) for $^1\Sigma$ states,

$$U^{\text{eff}}(z) = c_0 z^2(1 + \sum_{j=1} c_j z^j) + \sum_{j=1} m_e h_j^A z^j/M_A + \sum_{j=1} m_e h_j^B z^j/M_B$$
$$+ B_e(R_e/R)^2[J(J+1)][1 + \sum_{j=0} m_e g_j^A z^j/M_A + \sum_{j=0} m_e g_j^B z^j/M_B], \tag{7}$$

is in fact now identical in form to that of Eqs. (3)–(5), it appeared that the results obtained in Ref. (*13*) for LiH $X\,^1\Sigma^+$ could be employed to test the reliability of the algebraic approach using the algorithms developed in this laboratory for direct solution of the radial Schrödinger equation with an effective J-dependent potential. At the outset, Ogilvie's results were questionable since the reported value R_e = 1.595 141 37(186) Å, which can only be identified with R_e^{BO}, is clearly in disagreement with the previous determinations summarized above. Subsequently, however, it became known (*15*) that because of a programming error, all the parameters given in Ref. (*13*) are incorrect. Nevertheless, it has been established that although Ogilvie is successful in representing the precise multi-isotopomer data at low v for LiH $X\,^1\Sigma^+$ by a small set of parameters, the revised values being improved significantly over those available in Ref. (*13*), the functions defined by these parameters still lack an exact quantum mechanical significance. It is found that eigenvalues from the corresponding Hamiltonian for nuclear motion in LiH do not accurately represent the observed line positions; the magnitudes of the discrepancies between calculated and observed positions are systematically larger than the uncertainties of the data by up to two orders of magnitude. The present work, then, is concerned with a new determination of the radial form of the LiH $X\,^1\Sigma^+$ Hamiltonian, which is shown to be in accord with the experimental data to within the measurement errors.

RESULTS

The numerical procedure developed in this laboratory (*10*) was employed to determine new radial functions from the same data set as that considered by Ogilvie (*13*), except that in addition to rejection of the R(8) line in the 3–2 band of ^{7}LiH, several other lines with anomalously large residuals were also removed. (The rejection of these data, however, has only a minimal effect on the results obtained, and in no way accounts for any significant differences between the present results and those found by the method of Ref. (*13*)). Eigenvalue calculations were performed by numerical solution of the radial Schrödinger equation with effective potentials for each J value of the individual isotopomers; integrations were made over the range 0.9–2.8 Å, with a mesh size of 0.001 Å. Iteration of the eigenvalue calculations employed a convergence criterion of 0.000 001 cm^{-1}; the results were insensitive to a decrease in mesh size or an increase in the radial range.

Two fits were performed: the first fit was performed with a modified version of the computer program described in Ref. (*10*), and employed the same model, Eq. (7), as that described by Ogilvie (*13*), but with a slightly different set of parameters. The reduced standard deviation of the fit with 570 measured line positions and only 17 adjustable parameters was $\hat{\sigma} = 0.951$, indicating that on average, the measurements were represented to within their associated uncertainties. The fitted parameters are listed in Table I.

The second fit employed the same model as that of our earlier work and is described here since it provides convincing verification that results obtained in direct fits of radial Hamiltonians are essentially model independent. The Born–Oppenheimer potential is defined in terms of the variable-β Morse function,

$$U^{\mathrm{BO}}(R) = \mathcal{D}_e[1 - e^{-\beta(R)(R-R_e^{\mathrm{BO}})}]^2, \tag{8}$$

where

$$\beta(R) = \sum_{k=0} \beta_k (R - R_e^{\mathrm{BO}})^k. \tag{9}$$

The correction terms of Eqs. (4) and (5) are modeled as

TABLE I

Coefficients of Radial Functions[a] for the $X^1\Sigma^+$ State of LiH

	Fit to Eq. (7)		Fit[b] to Eqs. (8-11)
R_e/Å	1.594 911 19(85)	R_e/Å	1.594 910 45(85)
c_0/cm^{-1}	65723.8799(2074)	β_0	1.12856 9574(1804)
c_1	-0.8971 3744(1910)	β_1	-0.03438 7144(5801)
c_2	0.3482 0804(7052)	β_2	0.0641 4235(1993)
c_3	-0.089 6036(2682)	β_3	[0.0]
c_4	-0.03 9360(1087)	β_4	0.001 6504(1206)
c_5	-0.01 1160(2953)	β_5	0.002 4400(3907)
		β_6	-0.001 8781(5921)
h_1^{Li}/cm^{-1}	-7810.0(637.1)	u_1^{Li}	-2.8437(2193)
h_2^{Li}/cm^{-1}	9976.(2385.)	u_2^{Li}	2.8071(5224)
h_3^{Li}/cm^{-1}	-50552.(12200.)	u_3^{Li}	-4.3717(1.0190)
h_1^{H}/cm^{-1}	-102 250.5(182.6)	u_1^{H}	-35.18341(6108)
h_2^{H}/cm^{-1}	152 693.1(415.4)	u_2^{H}	43.6753(1329)
h_3^{H}/cm^{-1}	-190 063.(7142.)	u_3^{H}	-48.4396(8462)
h_4^{H}/cm^{-1}	210 955.(15 760.)	u_4^{H}	56.178(1.946)
		u_5^{H}	-57.762(6.762)
		u_6^{H}	33.055(7.006)
g_1^{Li}	0.4688(1592)	ρ_1^{Li}	0.0001 1998(5262)
g_1^{H}	0.69753(4208)	ρ_1^{H}	0.0002 0297(3064)
g_2^{H}	0.7369(2072)	ρ_2^{H}	0.0000 4045(2178)

[a]Integration of the radial Schrödinger equation was performed in the range 0.9 - 2.8 Å, with a mesh size of 0.001 Å.

[b]$\mathcal{D}_e$ = 20286.0 cm^{-1}; $\mathcal{R}$ = 1.00 Å; units of β_i, u_i and ρ_i are Å$^{-(n+1)}$, u cm^{-1} Å^{-n} and u Å^{-n}, respectively.

$$U_i(R) \equiv m_e\tilde{S}_i(R) = \sum_{k=1} u_k(R - R_e^{BO})^k, \tag{10}$$

and

$$m_e\tilde{R}_i(R) = \sum_{k=1} \rho_k[(R - \mathcal{R})^k - (R_e^{BO} - \mathcal{R})^k], \tag{11}$$

where the expansion pivot $\mathcal{R}$ is chosen as the smallest R value to which the data are sensitive. It is implicit in Eq. (11) that the functions adopt the arbitrary constraint $\tilde{R}_i(R_e^{BO}) = 0$. The fit performed with the model of Eqs. (8)–(11), with the same data set of 570 transitions, had a reduced standard deviation of $\hat{\sigma} = 0.946$, essentially the same as that of the first fit with the Dunham-like potential and the z-variable for the correction functions. The fitted parameters are listed in Table I, in which all parameters are given with more significant figures than is merited by the associated standard deviations in order that round-off error is negligible when using the parameters in calculations of effective potentials and eigenvalues.

DISCUSSION

Precisely known transition energies in the lower vibrational levels of the $X\,^1\Sigma^+$ state of ^{6}LiH, ^{7}LiH, ^{6}LiD, and ^{7}LiD have been employed in the present work in fits of the Born–Oppenheimer potential and associated radial functions describing deviation from Born–Oppenheimer behavior. The fits are achieved using numerical solution of the radial Schrödinger equation with effective potentials for each isotope and J value, and the quantum mechanical eigenvalues calculated from the derived functions reproduce the measured line positions to within the associated uncertainties.

It is of interest to compare the results of the fit performed using the same model as that of the formally equivalent algebraic approach described recently by Ogilvie (*13*) with those of the fit employing Eqs. (8)–(11). The most straightforward comparison is in the fitted R_e values; the values given in Table I, R_e = 1.594 911 2(9) Å and R_e = 1.594 910 4(9) Å, are in excellent agreement with each other as well as with other literature values (*2, 3, 5*) cited earlier.

Other comparisons are best made between the functions generated by the parameters in Table I. Figure 1 shows the function $\Delta U^{BO}(R) = U_\beta^{BO}(R) - U_z^{BO}(R)$, where $U_\beta^{BO}(R)$ and $U_z^{BO}(R)$ are the potentials generated by the sets of β_i and c_i in Table I. The magnitude of $\Delta U^{BO}(R)$ is extremely small ($\leqslant \sim 0.01$ cm^{-1}) over most of the radial range of Fig. 1. This demonstrates quite convincingly that the present procedure for obtaining a Born–Oppenheimer potential is essentially model-independent. The onset of larger discrepancies at the extremities of the radial range in Fig. 1 corresponds approximately to the classical turning points of the highest vibrational level in the data set, $v = 10$. It can be noted that the Born–Oppenheimer potential was modeled adequately in the present work using Eq. (7) with a smaller set of parameters (R_e, c_0–c_5) than in Ref. (*13*) (R_e, c_0–c_7), where the estimates of c_6 and c_7 appear to have anomalously large magnitudes.

There is similarly good correspondence for the $h^{Li}(R)$ and $h^H(R)$ functions, as shown in Fig. 2. Here again, as is clearly desirable, it is found that results are essentially unchanged between the different representations employed. Of particular interest is a consideration of the slopes of these functions at R_e, as given by the parameters h_1^{Li} and h_1^H. We have considered previously (*10*) the relationship between the slopes of the $\tilde{S}_{A,B}(R)$ functions and the R_e values of individual isotopomers. From Eq. (24) of Ref. (*10*),

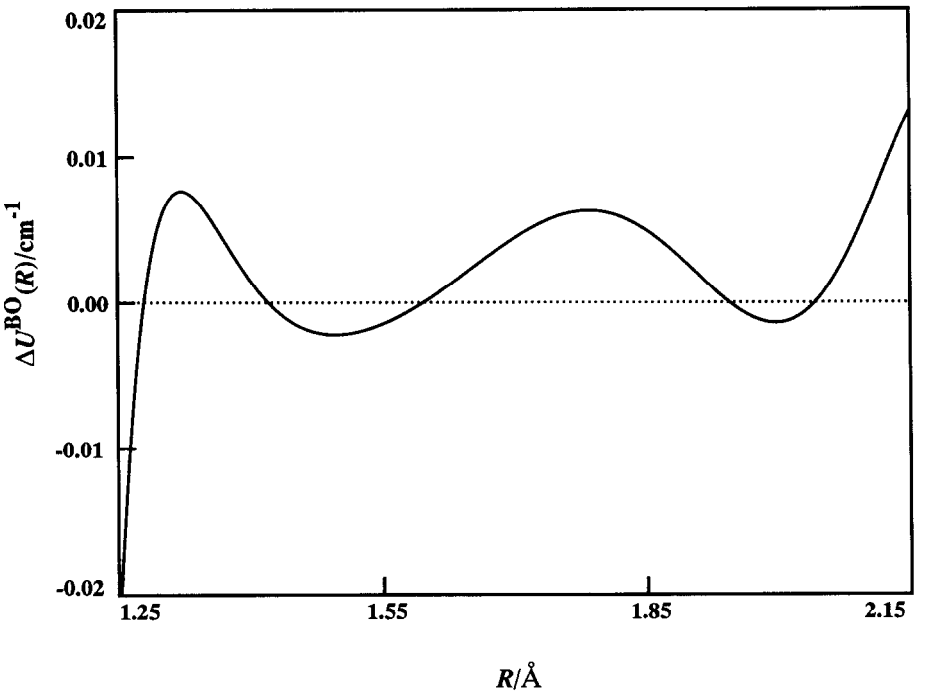


FIG. 1. Comparisons of estimated $U^{BO}(R)$ for LiH ($X^1\Sigma^+$). The function plotted is defined by $\Delta U^{BO}(R) = U_\beta^{BO}(R) - U_z^{BO}(R)$, where $U_\beta^{BO}(R)$ and $U_z^{BO}(R)$ are defined by the sets of β_i and c_i coefficients in Table I.

$$R_e^{QM} = R_e^{BO} - \frac{m_e}{k^{BO}}\left[\frac{1}{M_A}\left(\frac{d\tilde{S}_A(R)}{dR}\right)_{R_e^{BO}} + \frac{1}{M_B}\left(\frac{d\tilde{S}_B(R)}{dR}\right)_{R_e^{BO}}\right], \tag{12}$$

where k^{BO} is the force constant of the Born–Oppenheimer potential, we can obtain,

$$R_e^{QM}(^6\text{LiH}) - R_e^{QM}(^6\text{LiD}) = -\frac{m_e}{k^{BO}}\left(\frac{d\tilde{S}_H(R)}{dR}\right)_{R_e^{BO}}\left[\frac{1}{M_H} - \frac{1}{M_D}\right]. \tag{13}$$

Analogous equations can be written for $R_e^{QM}(^7\text{LiH}) - R_e^{QM}(^7\text{LiD})$, $R_e^{QM}(^6\text{LiH}) - R_e^{QM}(^7\text{LiH})$, and $R_e^{QM}(^6\text{LiD}) - R_e^{QM}(^7\text{LiD})$. Using the Y_{01} values of Ref. (*2*) to obtain relative R_e^{QM} values, the differences $R_e^{QM}(\text{LiH}) - R_e^{QM}(\text{LiD}) = 0.000\ 3364$ Å (for both ^{6}Li and ^{7}Li), and $R_e^{QM}(^6\text{LiH}) - R_e^{QM}(^7\text{LiH}) = 0.0000\ 0122$ Å (for both H and D) lead to $h_1^H = -1.020 \times 10^5$ cm^{-1} and $h_1^{Li} = -7.7 \times 10^3$ cm^{-1}. These values are in excellent agreement with the fitted values obtained in the present work (Table I), and clearly support the overall integrity of the results obtained.

The different estimates of $g^H(R)$ are shown in Fig. 3. Here again, there is rather good agreement between the two functions estimated in the present work from the g_i^H and ρ_i^H coefficients. As indicated earlier, the $g^H(R)$ function is determined with the arbitrary choice of $g^H(R_e) = 0$. In this context, it is appropriate to comment on the criticism in Ref. (*13*) of our "attempt" in similar work for HCl/DCl (*7*) to determine the radial function of Eq. (11) for the H center. This criticism was made despite our emphasis that the use of Watson's (*9*) arbitrary choice that $\tilde{R}_{A,B}(R_e)$ be set to zero, or equivalently the constraint $\rho_0^{A,B} = 0$ or $g_0^{A,B} = 0$, is not in conflict with

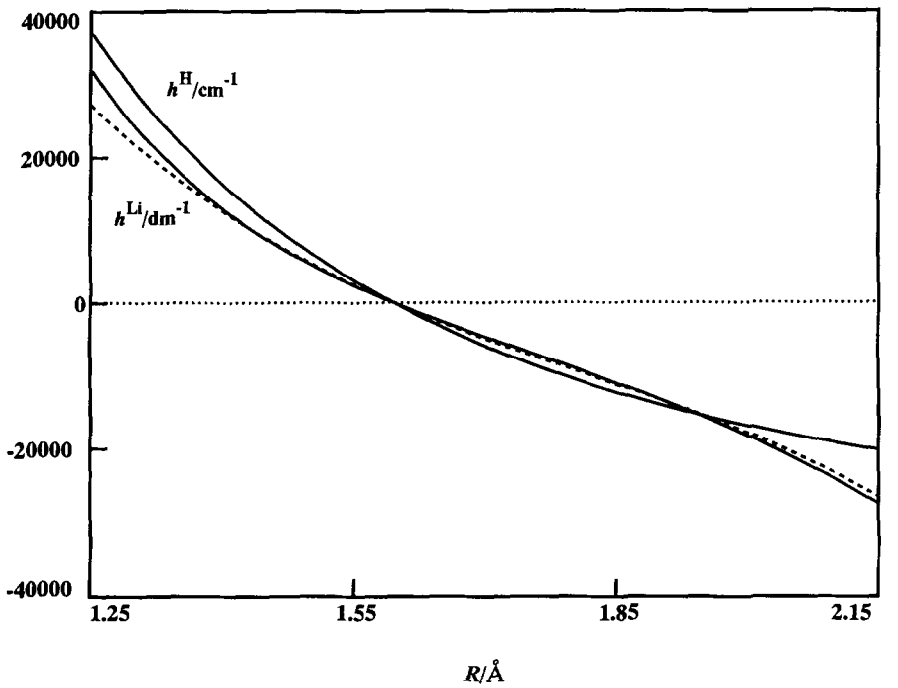


FIG. 2. The radial functions $h^{Li}(R)$ and $h^{H}(R)$ for LiH ($X\,^1\Sigma^+$). In each case, the two functions are defined by the parameters h_j (solid curve) and u_j (broken curve) in Table I. For $h^H(R)$, the two curves are coincident. The factor of 10 difference in scale between the $h^{Li}(R)$ and $h^H(R)$ functions should be noted.

the finite rotational magnetogyric ratio of HCl. It must be understood that the $\tilde{R}_{A,B}(R)$ functions, (as defined also by Ogilvie's $g_j^{A,B}$ parameters), cannot be identified with the purely heterogeneous $g_{rot}(R)$, but contain both heterogeneous and homogeneous contributions in an indeterminate proportion. There is a corresponding indeterminacy of the $U^{eff}(R)$ functions of Eq. (4) which analogously do not describe purely homogeneous mixing. In order to associate the $g_j^{A,B}$ coefficients of Eq. (7), (or the $\rho_j^{A,B}$ coefficients of Eq. (11)) with the experimental magnetogyric ratio, it would be necessary to adopt a different form for the effective Hamiltonian (*16*) in which the kinetic energy operator was defined as $[1 + g_{vib}(R)]\mathbf{P}_R^2/(2\mu_{nuc})$, rather than $\mathbf{P}_R^2/(2\mu_{at})$, $g_{vib}(R)$ describing purely homogeneous $^1\Sigma - X\,^1\Sigma$ non-adiabatic coupling.

The preceding discussion has focused on the generally excellent agreement between the radial functions obtained in the present work by numerical methods with different model forms. The radial variations shown in Figs. 2 and 3 are found to be similar to those determined by Ogilvie (*13, 15*) by an algebraic procedure. At the same time, however, it is essential to emphasize that the differences between the two approaches are highly significant. Since a detailed description of the method employed in Ref. (*13*) has not yet been published, no satisfactory explanation can be given for the failure of the quantum mechanical eigenvalues from Ogilvie's functions to reproduce the measured line positions to within the measurement accuracies, and hence also to provide reliable predictions for unobserved transitions. It is possible, however, that the algebraic method might suffer from the assumption that the contributions of $h(R)$ and $g(R)$ to the Z_{kl} are linearly independent. On the other hand, the present results are quantum-mechanically exact, and as we have shown for CO $X\,^1\Sigma^+$ (*17*), the derived

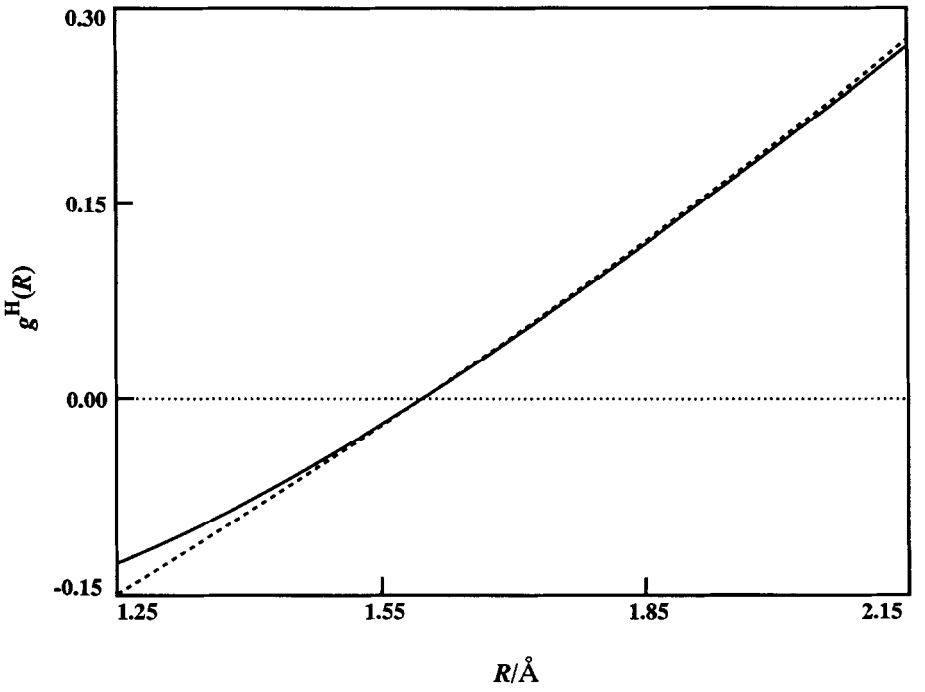


FIG. 3. The radial function $g^H(R)$ for LiH ($X^1\Sigma^+$). The functions are defined by the parameters g_j^H (broken curve) and ρ_j^H (solid curve), given in Table I.

functions can provide highly reliable predictions. To be of value, any compact representation of the type described in the present work requires this attribute. Numerical solution of the radial Schrödinger equation is now a routine operation in many laboratories so that the availability of a reliable Hamiltonian in radial form is of immediate practical value, more so than the availability of sets of Dunham coefficients. Any application with the radial forms determined by the method of Ref. (*13*) would require access to the complex computer code that can generate term values from the derived coefficients.

In conclusion, it should be noted that while the present work has provided results employing the model (*13*) of Eq. (7), this is necessary to facilitate a future comparison with the revised results of Ref. (*13*). The alternative fit using the model of Eqs. (8)–(11) is just as satisfactory; as shown in Figs. 2 and 3, the two sets of fitted functions describing deviations from Born–Oppenheimer behavior are in close agreement. Similarly, there is an excellent correspondence between the fitted $U^{BO}(R)$ potentials, with the two functions deviating from each other (Fig. 1) by less than 0.02 cm^{-1} over the range R = 1.26–2.26 Å. In general, however, the use of the variable-β Morse model is much preferred; $\beta(R)$ is a slowly varying function, even for R values corresponding to data close to dissociation, and can be modeled reliably by a relatively low order polynomial expansion. On the other hand, even with Ogilvie's expansion with the z variable, a large number of highly correlated coefficients are required with a Dunham-like potential when the available data extend beyond just a few low-lying vibrational levels.

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